

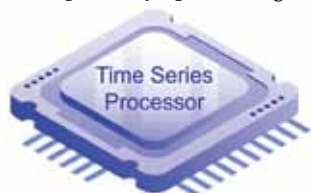
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Edge time series processor

The edge time series processor (TSP) from ABR can process large time series data with lower power. It is based on a new algorithm for AI signal processing called the provably optimal Legendre



Memory Unit (LMU) architecture. It can process time series including speech, language, audio, biosignals, RF signals, network traffic, and more at the edge. The TSP is capable of providing 10x to 25x cost advantage over CPU/GPU and 10x to 100x power advantage over existing algorithms computed on CPU/GPU. The edge time series processor SP can enable engineers to design devices that save power and increase response speed, accuracy, and privacy while lowering costs compared to most other existing solutions.

ABR

<https://appliedbrainresearch.com>

Smallest Wi-Fi HaLow SoC

AzureWave Technologies has released the world's smallest Wi-Fi HaLow module powered by Morse Micro MM6108 microchip. The module



measures just 13mm × 13mm and can deliver a 10x higher range compared to traditional Wi-Fi solutions. It runs

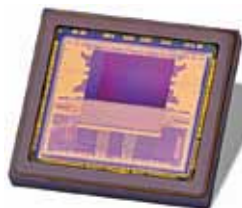
on lower energy and can deliver faster data than the existing solutions. This module can enable engineers to design smaller and energy-efficient IoT products. It can shorten the development process and time to market while providing a transmission range of up to 1km, which makes it suitable for use in IoT applications in large warehouses, smart buildings, smart farms, etc.

AzureWave Technologies

<https://www.azurewave.com>

Low-light 3D image sensor

Hydra3D+ is a Time-of-Flight (ToF) based CMOS image sensor from Teledyne e2v. The image sensor is capable of detecting and capturing 3D images in all light conditions. It offers a 832 × 600 pixel resolution and features a brand-new



10μm three-tap pixel, which provides very fast transfer times (starting from 10ns). The sensor can take images in extremely low light and without motion artefacts, making it suitable for pick and place, logistics, factory automation, factory safety, etc.

Teledyne e2v

<https://imaging.teledyne-e2v.com>

Wireless charger for multiple devices

AirFuel RF released by AirFuel Alliance is a new global interoperable standard for RF wireless power transfer. It uses radio waves to transfer energy wirelessly and can charge devices up to a few metres. The AirFuel RF can enable wireless charging and powering multiple devices simultaneously within

a three-dimensional area. Moreover, this RF technology can be embedded into very small electronic devices, such as electronic shelf labels, wearables,



hearable, IoT sensors, etc. It can enable engineers to eliminate battery and design a truly wireless electronics, thus making the device more compact, lighter, sturdier, and cheaper.

AirFuel

<https://airfuel.org>

Bluetooth sensor with energy harvesting

SMK's Gen 2 SCPS is a Bluetooth based ambient sensor with energy harvesting capabilities. The Gen 2 SCPS sensor can collect ambient information such as temperature, humidity, luminance, carbon-dioxide, air pressure, accelerometer value, and ambient sound. The energy harvesting capability of Gen 2 SCPS sensor can allow for a battery-free operation of low-powered electronics. The energy harvesting capability and multi-sensing capability in a single package will enable the engineers to develop a light weight, compact, and no maintenance sensor node or other IoT devices that can communicate using Bluetooth.

SMK

<https://www.smkusa.com>

Low-powered Wi-R solution

The Ixana YR11 chip is equipped with a new Wi-R protocol which consumes up to 100x lower energy compared to Bluetooth or existing Wi-Fi solutions. The solution employs a non-radiative